

Web User Interface (HCI) Design & Development

Presentation 2 — Front-End Web Development

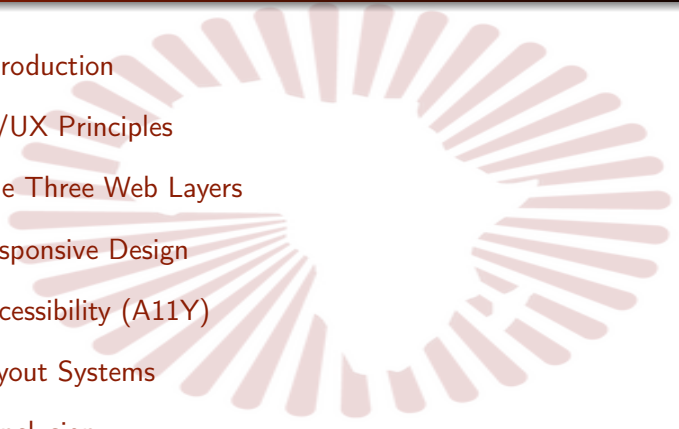
Wandiya James
AKAKPO A. A. Sara
Mairame Samba Niang
Ndeye Binta Ndiaye
Ndeye Khady Wade
Eric Judicael Akoto

African Institute for Mathematical Sciences (AIMS-Senegal)

Supervised by Osias Noel N. F. Tossou

May 5, 2026

Overview

- 
- 1 Introduction
 - 2 UI/UX Principles
 - 3 The Three Web Layers
 - 4 Responsive Design
 - 5 Accessibility (A11Y)
 - 6 Layout Systems
 - 7 Conclusion



Introduction

introduction

In an ever-changing digital world, designing intuitive and high-performing web user interfaces is essential. This presentation explores the fundamentals of UI/UX design and the key technologies needed to create exceptional web experiences.



AIMS

African Institute for
Mathematical Sciences
SENEGAL

What is a Web Interface?

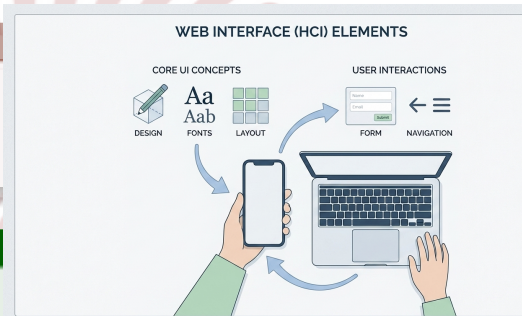
The front-end layer users interact with

Definition — HCI

Web UI encompasses all visual and interactive elements a user sees: layouts, typography, navigation, forms, and animations.

Why it Matters

- **53%** users leave if load time > 3s.
- **88%** won't return after a bad UX.
- **1B+** people live with disabilities.
- **100ms** is the limit for "instant" feel.



AIMS

African Institute for
Mathematical Sciences
SENEGAL

Design Principles

Foundations for Effective Interfaces

1. Visual Hierarchy

Use **size**, **color**, and **contrast** to guide the eye toward key actions.

2. Consistency

Uniform components and patterns reduce cognitive load and build trust.

3. Simplicity

Remove fluff. Use white space and clear Call-to-Actions ().

4. Feedback

Every action must produce an immediate visual response (<100ms).

5. Accessibility

Design for ALL users, including those with disabilities. Minimum contrast ratio, keyboard navigation, and ARIA labels are required.

6. Responsive Design

The interface must **adapt** gracefully to any screen size, from a 320px mobile to a 4K desktop monitor.



AIMS

African Institute for
Mathematical Sciences
SENEGAL

The Three-Layer Architecture

Strict Separation of Concerns

Layer	Technology	Role
Structure	HTML5	Semantic content & A11Y tree
Presentation	CSS3	Styling, layout, responsiveness
Behavior	JS ES6+	Interactivity & Async data

Best Practice

Keep them separate! Inline styles or layout logic in JS makes code hard to maintain and test.



HTML5 — The Structure Layer

Semantic elements, multimedia and accessibility

elements and accessibility :

- `<header>`, `<nav>`, `<main>`
- `<section>`, `<article>`, `<aside>`
- `<footer>`, `<figure>`,
`<figcaption>`

New input types :

- date, email, range, color

Multimedia & A11Y :

- `<video>`, `<audio>`, `<canvas>`
- ARIA Roles & attributes data-*

HTML5 structure

```
<html lang="fr">
<body>
<header>
<nav aria-label="Nav principale">
<a href="/">Accueil</a>
</nav>
</header>
<main id="contenu-principal">
<article>
<h1>Titre de page</h1>
</article>
</main>
<footer>...</footer>
</body>
```



AIMS

African Institute for
Mathematical Sciences
SENEGAL

CSS3 — Presentation Layer

Layout, theming, animation, responsive

Layout systems:

- Flexbox — 1-D layout
- CSS Grid — 2-D layout

Theming & design tokens:

- Custom properties (`var()`)
- Variables in `:root`

Animation & Responsive:

- CSS transitions & animations
- Media queries
- `clamp()` fluid typography

CSS3 — Design tokens

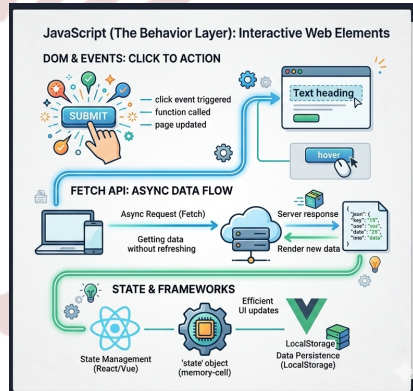
```
:root {  
  -primary-color: #065A82;  
  -accent-color: #02C39A;  
  -radius: 8px;  
}  
  
.card-grid {  
  display: grid;  
  grid-template-columns:  
    repeat(auto-fit,  
      minmax(280px, 1fr));  
  gap: 1.5rem;  
}  
  
.button:hover {  
  background: var(-accent-color);  
  transition: background 0.2s ease;  
}
```

JavaScript — The Behavior Layer

DOM Manipulation and Async data

Capabilities:

- **DOM Events:** Click, hover, submit.
- **Fetch API:** Getting data without refreshing.
- **State Mgmt:** LocalStorage & Frameworks (React/Vue).

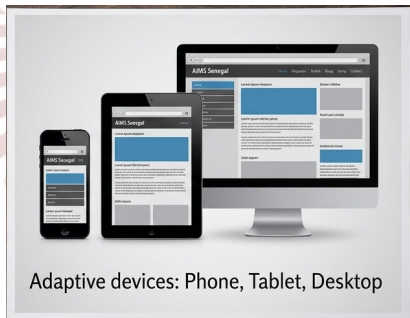


Responsive Web Design (RWD)

Adaptability for all screen sizes

Concept

Interfaces that adapt fluidly to any screen size (Mobile, Tablet, Desktop) using a single codebase.



Adaptive devices: Phone, Tablet, Desktop

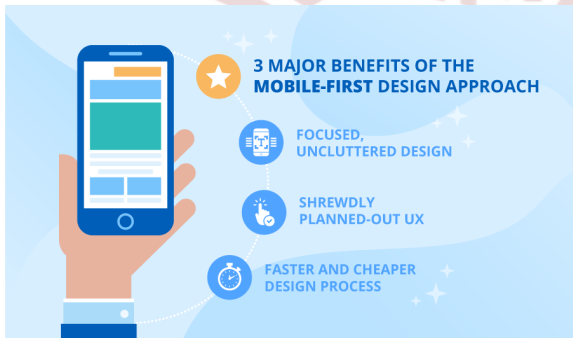


AIMS

African Institute for
Mathematical Sciences
SENEGAL

Mobile-First Strategy

Designing for the smallest screen first



Why Mobile-First?

- Focuses on essential content.
- Better performance on low bandwidth.
- Progressive enhancement vs. Graceful degradation.



AIMS

African Institute for
Mathematical Sciences
SENEGAL

Web Accessibility (A11Y)

WCAG 2.1 — Making the web inclusive

A Legal and Moral Duty

1.3 Billion people live with disabilities. A11Y is not a feature; it's a requirement for EU/US compliance.

The POUR Principles:

- **Perceivable:** Info must be visible to all senses.
- **Operable:** Interface cannot require a mouse.
- **Understandable:** Users must know how to use it.
- **Robust:** Must work with Screen Readers.



Practical A11Y Checklist

- **Alt Text:** For every meaningful image.
- **Contrast:** Minimum 4.5:1 ratio.
- **Keyboard:** Visible focus indicators (:focus).
- **Labels:** Every input must have a visible `<label>`.
- **Skip Links:** "Skip to Main Content" for keyboard users.



CSS Flexbox — One-Dimensional Layout

Ideal for: navigation bars, card rows, centering

Key Properties

- `display: flex`
- `justify-content`
space-between, center, flex-start
- `align-items`
center, stretch, flex-end
- `flex-wrap` : multi-line
- `gap` : spacing between items

Visual Result

`flex-direction: row`

Item 1

Item 2

Item 3

Flexbox Code

```
.container {  
  display: flex;  
  justify-content: space-between;  
  align-items: center;  
  gap: 1rem;  
}  
  
/* Auto-wrapping cards */  
.card-row {  
  display: flex;  
  flex-wrap: wrap;  
  gap: 1.5rem;  
}  
  
.card-row > .card {  
  flex: 1 1 280px;  
  max-width: 360px;  
}
```



AIMS

Institute for
Mathematical Sciences
SENEGAL

CSS Grid — Two-Dimensional Layout

Ideal for: page structure, dashboards, galleries

Key Properties

- `display: grid`
- `grid-template-columns`
- `grid-template-areas`
- `gap`
- `auto-fit + minmax()`

Flexbox vs Grid

Flexbox	Grid
1 dimension	2 dimensions
Content	Layout
Components	Structure

Practical Rule

Use Flexbox for 1-D (row or column). Grid for 2-D (row and column). Freely combined.

Grid with Named Areas

```
.page {  
  display: grid;  
  grid-template-columns:  
    260px 1fr;  
  grid-template-rows:  
    64px 1fr 56px;  
  grid-template-areas:  
    "header header "  
    "sidebar content "  
    "footer footer ";  
  min-height: 100vh;  
}  
  
.header { grid-area: header; }  
.sidebar { grid-area: sidebar; }  
.content { grid-area: content; }
```



AIMS

African Institute for
Mathematical Sciences
SENEGAL

Key Takeaways

What to remember from this presentation

- **01 - Design for the human.** Every UI decision must serve the user. Clear hierarchy, intuitive flows and fast feedback create a successful experience.
- **02 - The 3 layers are inseparable.** HTML for structure, CSS for style, JavaScript for behaviour well separated, well organised.
- **03 - Responsive by default.** Design mobile-first. Fluid grids, flexible media and media queries to adapt to any screen.
- **04 - Accessibility is not optional.** WCAG principles ensure an interface usable by everyone, including the 1 B+ people living with a disability.

*“Good UI/UX is not a feature — it is **the product itself**”*



AIMS

African Institute for
Mathematical Sciences
SENEGAL

Conclusion

- **User-Centric Design:** UI is about the human, not just the code.
- **Layered Harmony:** HTML, CSS, and JS must work separately but together.
- **Inclusivity:** Responsive and Accessible design are the standards of the modern web.
- **Evolution:** Performance optimization is key to retaining users in a mobile-fast world.

Front-end development is the bridge between technology and human experience.



AIMS

African Institute for
Mathematical Sciences
SENEGAL

References

- 
-  E. Marcotte, *Responsive Web Design*, A List Apart, 2010.
 -  J. Nielsen, *10 Usability Heuristics*, NN Group, 1994.
 -  W3C, *Web Content Accessibility Guidelines (WCAG) 2.1*, 2018.
 -  Google, *Mobile Page Speed Benchmarks*, 2018.



Questions & Discussion

Thank you for your attention !

Questions & Discussion



AIMS

African Institute for
Mathematical Sciences
SENEGAL